

Reading Notes on *The Renormalized Volume of a 4-Dimensional Ricci-Flat ALE Space*

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Abstract

This note focuses on Section 3 of the paper by Biquard–Hein on the renormalized volume of four-dimensional Ricci-flat ALE spaces. For a Ricci-flat ALE space asymptotic to the flat cone $\mathbb{R}^4/\Gamma$, the paper defines a renormalized volume using the canonical CMC foliation of the end and proves that it is always nonpositive, with equality only in the flat case. We concentrate on this definition and on the proof of the inequality, recalling the asymptotic expansion of the metric and the classification of its leading term only as needed. The main argument is organized around a function u satisfying $\Delta u = 8$ and a prescribed asymptotic expansion: the renormalized volume is identified with the coefficient of the r^{-2} term, and the sign is derived from an integration-by-parts identity.

1 Introduction

Let (M^4, g) be a Ricci-flat ALE space asymptotic to $(\mathbb{R}^4/\Gamma, g_0)$, where $\Gamma \subset SO(4)$ is finite and acts freely on $\mathbb{S}^3$. By the basic ALE theory, one has a diffeomorphism at infinity

$$\Phi : (\mathbb{R}^4 \setminus B_1(0))/\Gamma \longrightarrow M \setminus K \quad (1)$$

such that for every $k \geq 0$,

$$|\nabla_{g_0}^k (\Phi^* g - g_0)|_{g_0} = \mathcal{O}(r^{-4-k}) \quad (2)$$

as $r \rightarrow \infty$, where g_0 is the Euclidean metric. More precise expansions in suitable gauges are also available; see [10].

The paper of Biquard–Hein [4] proves two statements which should be understood together.

- (1) One can define a renormalized volume for (M, g) by comparing the volume enclosed by the canonical CMC leaves with the volume of the cone balls $B_\rho(0)/\Gamma$.
- (2) This renormalized volume is always nonpositive, and it vanishes only in the flat case.

This two statements should be viewed as parts of a single geometric picture. The inequality $\mathcal{V}(g) \leq 0$ is the final conclusion, but before discussing the proof, it is worth explaining why the volume difference above is the natural quantity to consider in the Ricci-flat ALE setting. This is easier to understand in the broader context of how one measures the deviation of a noncompact manifold from its model geometry at infinity. In geometric analysis, such invariants are typically extracted either from the first nontrivial term in the asymptotic expansion of the metric, or from the comparison of large geometric quantities with those of a model space. The present paper belongs to the second class, with the flat cone $\mathbb{R}^4/\Gamma$ as the asymptotic model.

To make this point more concrete, let us briefly recall several familiar examples of these two types of invariants. We begin with the classical mass-type invariant in asymptotically flat geometry, and then turn to volume-based constructions in asymptotically hyperbolic and related settings. Seen from this perspective, the Biquard–Hein renormalized volume fits naturally into the second circle of ideas, although the asymptotic model and the geometric mechanism used to define the comparison regions are specific to the ALE setting.

The standard example of the first type is the ADM mass. If (M^n, g) is asymptotically flat and

$$g_{ij} = \delta_{ij} + O(r^{2-n}),$$

then the ADM mass is defined by

$$m_{\text{ADM}}(g) = \lim_{r \rightarrow \infty} \frac{1}{2(n-1)\omega_{n-1}} \int_{S_r} (\partial_i g_{ij} - \partial_j g_{ii}) \nu^j d\sigma.$$

This is a flux integral at infinity, built from the leading deviation of g from the Euclidean metric. Under suitable scalar-curvature assumptions, the positive mass theorem asserts that

$$m_{\text{ADM}}(g) \geq 0,$$

with equality only for the Euclidean metric.

A basic example of the second type arises in the asymptotically hyperbolic Einstein setting. Let (M^4, g) be a complete asymptotically hyperbolic Einstein manifold, and let ρ be a defining function. One considers the compact domains $B(r) = \{\log \rho^{-1} \leq r\}$, then the volume admits an asymptotic expansion of the form

$$\text{Vol}_g(B(r)) = v_0 e^{3r} + v_1 e^r + V + o(1), \quad r \rightarrow \infty,$$

where the coefficients v_0 and v_1 depend on the choice of defining function, while the constant term V does not. The quantity V is called the renormalized volume. Thus the leading divergent terms are not intrinsic, but the constant term is. This is the classical notion of renormalized volume for asymptotically hyperbolic Einstein metrics; see Graham [12] and Anderson [1].

A related idea is to compare the volume of a large region directly with that of the corresponding region in a model space. Schematically, one studies a difference of the form

$$\text{Vol}_g(\Omega_\rho) - \text{Vol}_{\bar{g}}(\bar{\Omega}_\rho),$$

where $(\bar{M}, \bar{g})$ is the reference geometry and $\bar{\Omega}_\rho$ is the corresponding model region. The main point is then to understand the asymptotic behavior of this difference as $\rho \rightarrow \infty$. In favorable situations, this leads either to a finite renormalized quantity or to a comparison inequality. This point of view appears, for example, in volume comparison results for asymptotically hyperbolic manifolds; see Brendle–Chodosh [5] and Hu–Ji–Shi [13]. In a somewhat different direction, Huisken introduced the isoperimetric mass, which is defined by comparing the area–volume relation of large regions in an asymptotically flat manifold with the corresponding Euclidean relation; see Huisken [14]. Its relation to the ADM mass was studied in later work using Huisken’s isoperimetric mass; see, for instance, Jauregui–Lee [15]. Related results on isoperimetry, scalar curvature, and mass were obtained by Chodosh–Eichmair–Shi–Yu [9].

The construction of Biquard–Hein [4] belongs to this general circle of volume-type ideas, but its geometric setting is different. Here the asymptotic model is neither Euclidean space nor hyperbolic space, but the flat cone $\mathbb{R}^4/\Gamma$. The relevant comparison is therefore between large regions in (M, g) and the corresponding cone balls in $(\mathbb{R}^4/\Gamma, g_0)$.

At this point one immediately faces a geometric issue: which large regions should be used in the comparison? On a general ALE end, ordinary coordinate balls depend on the chosen asymptotic chart and so do not lead, by themselves, to an obviously intrinsic quantity. The canonical CMC foliation resolves this difficulty by providing a distinguished family of large hypersurfaces, and hence an essentially canonical exhaustion of the end.

More precisely, let Σ_ρ denote the canonical CMC leaf with mean curvature $3/\rho$, and let Ω_ρ be the region enclosed by Σ_ρ . One then defines

$$\mathcal{V}(g) = \lim_{\rho \rightarrow \infty} \left(\text{Vol}_g(\Omega_\rho) - \text{Vol}_{g_0}(B_\rho(0)/\Gamma) \right), \quad (3)$$

provided the limit exists. The main result of the paper is that this limit does exist and satisfies

$$\mathcal{V}(g) \leq 0, \quad (4)$$

with equality if and only if (M, g) is isometric to the flat cone $(\mathbb{R}^4/\Gamma, g_0)$.

This explains why the CMC foliation is not an auxiliary ingredient but part of the definition itself. It is precisely what makes the volume comparison geometric rather than coordinate-dependent.

This foliation is not only geometrically natural but also closely tied to the analytic structure of the proof. In the special asymptotic gauge used by Biquard–Hein, the leading term h_0 of the metric expansion satisfies

$$\partial_r \lrcorner h_0 = 0, \quad \mathrm{tr}_{g_0} h_0 = 0, \quad \mathrm{div}_{g_0} h_0 = 0.$$

These cancellations mean that the large coordinate spheres are already much closer to having constant mean curvature than the general ALE expansion alone would suggest: the first possible error term disappears, so the deviation becomes one order smaller. Because of this, one can modify the asymptotic chart so that its coordinate spheres agree with the canonical CMC leaves, while keeping the leading asymptotic term of the metric unchanged. This is why the CMC foliation is not introduced merely for convenience. It is what allows the volume comparison to be defined in a geometric, rather than coordinate-dependent, way.

There is also a deeper reason that makes this construction natural. The asymptotic model $(\mathbb{R}^4/\Gamma, g_0)$ carries an exact scaling structure. The radial dilations $x \mapsto \lambda x$ are generated by the vector field $r\partial_r$, and they satisfy

$$\mathcal{L}_{r\partial_r} g_0 = 2g_0.$$

More generally, if (N, h) is a Riemannian manifold, we shall say that a vector field X is *homothetic* if

$$\mathcal{L}_X h = 2c h$$

for some constant c . Equivalently, the local flow of X rescales the metric by constant factors. In this language, $r\partial_r$ is the basic homothetic vector field on the flat cone.

The scalar function naturally associated with this scaling field is r^2 . Indeed, on the flat cone one has

$$\nabla_{g_0}(r^2) = 2r\partial_r, \quad \Delta_{g_0}(r^2) = 8, \quad \mathrm{Hess}_{g_0}^0(r^2) = 0.$$

Thus r^2 is the potential for the radial homothety, and the vanishing of its trace-free Hessian expresses the fact that the cone scaling structure is exact. This is the basic feature of the model geometry that drives the later argument.

From this point of view, the main problem is no longer just to compare two volumes, but to understand whether some remnant of this cone scaling structure survives on a Ricci-flat ALE space, and if not, how its failure is recorded at infinity. The paper answers this by introducing a function u on (M, g) satisfying $\Delta_g u = 8$ and asymptotic to r^2 at infinity. The first decaying correction term in the expansion of u turns out to encode the renormalized volume, while the tensor $\mathrm{Hess}_g^0 u$ measures the failure of ∇u to generate a homothety. In this sense, renormalized volume should not be viewed as merely the limit of a volume difference. It is a scalar quantity that records how the exact scaling structure of the cone fails on the ALE space.

With this picture in mind, the structure of the proof becomes quite natural. One first puts the geometry at infinity into the right form, so that the comparison with the cone is made using canonically defined large regions rather than arbitrary coordinate spheres. One then studies a function u that plays, on the ALE space, the role played by r^2 on the cone, and shows that the renormalized volume is encoded in its asymptotic behavior. Finally, one extracts the sign from a global identity involving $\mathrm{Hess}_g^0 u$, which measures the failure of the cone scaling picture to persist exactly on (M, g) .

This is the perspective taken in the present note. Our aim is not only to explain why $\mathcal{V}(g) \leq 0$, but also to clarify the geometric idea behind the argument: the role of the CMC foliation in making the comparison canonical, the role of the cone scaling as the guiding model, and the role of the function u in linking the volume comparison to an interior analytic identity.

2 Asymptotic structure of the metric at infinity

To analyze the metric near infinity, one first passes from an arbitrary ALE chart to Bianchi gauge relative to g_0 . In this gauge the Ricci-flat equation becomes elliptic, and the metric admits an asymptotic expansion. For the discussion below, the leading term is the important one:

$$\Phi^* g - g_0 = h_0 + h', \tag{5}$$

where h_0 is homogeneous of degree -4 and satisfies

$$\mathcal{L}_{r\partial_r} h_0 = -2h_0, \quad B_{g_0} h_0 = 0, \quad \Delta_{g_0} h_0 = 0, \quad (6)$$

while the remainder obeys

$$\sum_{k=0}^{k_0} r^k |\nabla_{g_0}^k h'|_{g_0} = \mathcal{O}(r^{-5}).$$

Thus the first nontrivial asymptotic information is encoded in a degree -4 symmetric 2-tensor satisfying the linearized Ricci-flat and Bianchi-gauge equations.

The leading term h_0 contains both gauge and geometric contributions. The gauge part comes from infinitesimal changes of coordinates, and is represented by tensors of the form

$$\mathcal{L}_X g_0,$$

where X is a harmonic vector field on $\mathbb{R}^4 \setminus \{0\}$ of the form

$$X(x) = \frac{1}{r^4} Lx, \quad L \in \mathbb{R}^{4 \times 4}.$$

These tensors are homogeneous of degree -4 and satisfy the same linearized equations as h_0 , so they should be regarded as pure gauge terms. What remains after removing them is the genuinely geometric part. In Kronheimer's examples, this remaining part is what the paper calls a *reduced Kronheimer term*; representation-theoretically, it lies in the two irreducible components $S_4^\pm$.

Accordingly, every h_0 satisfying (6) admits a unique decomposition

$$h_0 = h_0^+ + h_0^- + \mathcal{L}_{X_1} g_0 + \mathcal{L}_{X_2} g_0 + \mathcal{L}_{X_3} g_0,$$

where $h_0^\pm$ are reduced Kronheimer terms and the X_i are harmonic vector fields of the allowed types.

This decomposition makes it possible to separate the geometric part of the leading term from the pure gauge part. After composing the ALE chart with the flows of the relevant harmonic vector fields, one may arrange that

$$\partial_r \lrcorner h_0 = 0, \quad \text{tr}_{g_0} h_0 = 0, \quad \text{div}_{g_0} h_0 = 0. \quad (7)$$

These identities give a particularly convenient normalized form of the leading asymptotic term. In particular, the trace-free condition controls the leading behavior of the volume form, the radial condition improves the comparison with large CMC hypersurfaces, and the divergence-free condition simplifies the boundary expressions that appear later. For this reason, it is this normalized form of h_0 that will be used from now on.

3 The renormalized volume inequality

3.1 The main idea

The proof of the inequality

$$\mathcal{V}(M, g) \leq 0$$

is easiest to understand by first looking at the flat cone model $(\mathbb{R}^4/\Gamma, g_0)$. On this model space, the function r^2 satisfies

$$\Delta_{g_0}(r^2) = 8, \quad \text{Hess}_{g_0}^0(r^2) = 0.$$

Moreover,

$$\nabla(r^2) = 2r\partial_r,$$

so r^2 is the potential of the radial vector field generating the natural scaling of the cone. This is the model for the whole argument.

On a general Ricci-flat ALE space (M, g) , the proof begins by putting the geometry at infinity into a canonical form. The renormalized volume is defined using the canonical CMC foliation, not arbitrary coordinate spheres. Thus one first modifies the asymptotic chart so that the coordinate spheres coincide with the CMC leaves Σ_ρ . This produces a CMC gauge at infinity and makes the definition of the renormalized volume geometric.

Only after this normalization does one introduce a function u that plays the role of r^2 on the flat cone. One looks for a solution of

$$\Delta_g u = 8$$

with the same leading behavior as r^2 at infinity. If the metric were exactly conical, one would expect $\text{Hess}^0 u$ to vanish, just as $\text{Hess}_{g_0}^0(r^2)$ does on the model cone. Thus $\text{Hess}^0 u$ measures the failure of the cone scaling structure to persist on (M, g) .

With this perspective, the proof has three main steps.

Step 1. First put the metric into the good asymptotic gauge and then modify the asymptotic chart so that

$$\Phi(\partial B_\rho(0)/\Gamma) = \Sigma_\rho$$

for all large ρ . This gives a CMC gauge at infinity and makes the definition of the renormalized volume geometric.

Step 2. Next solve

$$\Delta_g u = 8$$

with asymptotic expansion

$$u = r^2 + br^{-2} + \mathcal{O}_\varepsilon(r^{-3+\varepsilon}).$$

The coefficient b is the first nontrivial correction to the cone model, and it will be related to the renormalized volume.

Step 3. Finally, use an integral identity involving $\text{Hess}^0 u$ to prove that $b \geq 0$, with equality only in the flat case. This yields

$$\mathcal{V}(M, g) \leq 0.$$

In short, the proof has two layers. First, one uses the CMC foliation to normalize the geometry at infinity and define the renormalized volume geometrically. Second, one studies a special solution of $\Delta_g u = 8$, whose r^{-2} -coefficient is both related to the volume defect and controlled by an integral identity involving $\text{Hess}^0 u$.

3.2 Constructing the CMC gauge and the existence of the renormalized volume

The main point of this step *is not to* construct the CMC foliation itself, which is already available for large ρ , but to choose an asymptotic chart in which the coordinate spheres agree with those canonical CMC leaves. More precisely, starting from the asymptotic gauge produced in Section 2, we want to replace the chart Φ by a new one satisfying

$$\Phi(\partial B_\rho(0)/\Gamma) = \Sigma_\rho$$

for all sufficiently large ρ , where Σ_ρ denotes the CMC leaf of mean curvature $3/\rho$. The key issue is that this change of chart must not alter the coefficient h_0 of order r^{-4} in the metric expansion. This is exactly what one should expect: the CMC foliation is a geometric object, so passing from the original asymptotic

chart to a chart adapted to the CMC leaves should only amount to a higher-order normalization of the end, not to a change in the leading asymptotic data.

We now explain why this is true. Start from the asymptotic gauge, so that (5) and (7) hold. For $\rho \gg 1$, let $h_\rho(x) = \rho x$ and define the rescaled metric

$$g_\rho := h_\rho^* \Phi^* g = g_0 + \rho^{-4} h_0 + \mathcal{O}(\rho^{-5}) \quad (8)$$

on every fixed compact subset of $(\mathbb{R}^4 \setminus \{0\})/\Gamma$. Thus, after rescaling, the metric is a small perturbation of the flat cone metric g_0 , and the problem becomes one of finding a CMC hypersurface close to the unit sphere S^3/Γ .

To parametrize such a nearby hypersurface, choose a small function $f_\rho : S^3/\Gamma \rightarrow \mathbb{R}$ and consider the radial graph

$$\tilde{\Sigma}_\rho = \{(1 + f_\rho(x))x : x \in S^3/\Gamma\}.$$

Here $f_\rho(x)$ measures the change of radius in the direction x . One asks that $\tilde{\Sigma}_\rho$ have constant mean curvature 3 with respect to the metric g_ρ . The linearization of this equation at the round sphere is

$$J = \Delta_{S^3/\Gamma} + 3.$$

When $\Gamma \neq \{1\}$, the translation modes present on S^3 disappear on S^3/Γ , so J is invertible. Therefore, by the implicit function theorem, for all sufficiently large ρ there exists a unique small function f_ρ such that $\tilde{\Sigma}_\rho$ has constant mean curvature 3 in the metric g_ρ . Pushing $\tilde{\Sigma}_\rho$ forward by $\Phi \circ h_\rho$ gives the corresponding CMC leaf $\Sigma_\rho \subset M$.

At this point the CMC leaves have already been constructed. The next question is how close they are to the coordinate spheres $\Phi(\partial B_\rho(0)/\Gamma)$, because this determines whether one can replace the original asymptotic chart by a new one for which the coordinate spheres coincide with the CMC leaves, without changing the leading term h_0 .

If one only uses the rough estimate (8), then the mean curvature of the unit sphere S^3/Γ in the rescaled metric differs from 3 by $\mathcal{O}(\rho^{-4})$. Since the linearized operator is invertible, the implicit function theorem yields $f_\rho = \mathcal{O}(\rho^{-4})$ in the rescaled picture. Here f_ρ measures the radial correction of the unit sphere, so after returning to the original scale the corresponding correction along the sphere of radius ρ has size $\rho f_\rho = \mathcal{O}(\rho^{-3})$. Thus the standard argument would only show that, in the original scale, the CMC leaf can be obtained from the coordinate sphere by moving each point along the normal direction by an amount of order $\mathcal{O}(\rho^{-3})$.

The present situation has its peculiarities. We have constructed the good gauge conditions:

$$\partial_r \lrcorner h_0 = 0, \quad \text{tr}_{g_0} h_0 = 0, \quad \text{div}_{g_0} h_0 = 0$$

force an additional cancellation in the expansion of the mean curvature of the coordinate spheres. As a result, the mean-curvature error improves from $\mathcal{O}(\rho^{-4})$ to $\mathcal{O}(\rho^{-5})$. Repeating the same perturbative argument, one gains one more order and obtains $f_\rho = \mathcal{O}(\rho^{-5})$ in the rescaled picture. Returning to the original scale, this means that the CMC leaf Σ_ρ is a normal graph over the coordinate sphere $\Phi(\partial B_\rho(0)/\Gamma)$ with graph function of size $\mathcal{O}(\rho^{-4})$.

This $\mathcal{O}(\rho^{-4})$ size is exactly what is needed. It shows that one can move the coordinate sphere onto the CMC leaf by a correction that is one order smaller than the leading metric term h_0 . Concretely, define a map F on the end by

$$F(\rho x) := (1 + f_\rho(x))\rho x, \quad \rho \gg 1, \quad x \in S^3/\Gamma,$$

and then define the new asymptotic chart by

$$\Psi := \Phi \circ F.$$

By construction,

$$\Psi(\partial B_\rho(0)/\Gamma) = \Phi(F(\partial B_\rho(0)/\Gamma)) = \Sigma_\rho.$$

Thus Ψ is precisely the desired chart adapted to the CMC foliation.

It remains to check that passing from Φ to $\Psi = \Phi \circ F$ does not change the coefficient h_0 . Write

$$F = \text{Id} + V.$$

Since $f_\rho = \mathcal{O}(\rho^{-5})$ in the rescaled picture, the corresponding normal displacement in the original scale has size $\rho f_\rho = \mathcal{O}(\rho^{-4}) = \mathcal{O}(r^{-4})$, and therefore $V = \mathcal{O}(r^{-4})$, $\nabla V = \mathcal{O}(r^{-5})$. Directly computations reads, if we decompose

$$\Psi^*g = F^*(\Phi^*g) = F^*g_0 + F^*h_0 + F^*\mathcal{O}(r^{-5}),$$

then

$$\Psi^*g = g_0 + h_0 + \mathcal{O}(r^{-5}).$$

Therefore the coefficient of order r^{-4} is unchanged. *In particular, after renaming Ψ as Φ , we may assume that*

$$\Phi(\partial B_\rho(0)/\Gamma) = \Sigma_\rho$$

for all large ρ , while the leading term h_0 remains the same.

Once this CMC gauge is in place, the renormalized volume is defined as the limit of the following function

$$g(\rho) := \text{Vol}_g(\Omega_\rho) - \text{Vol}_{g_0}(B_\rho(0)/\Gamma).$$

Then we show that this is well defined.

Write

$$\Phi^*g = g_0 + h_0 + h',$$

where $h_0 = \mathcal{O}(r^{-4})$, $h' = \mathcal{O}(r^{-5})$, and $\text{tr}_{g_0} h_0 = 0$. Then one consequence of the property $\text{tr}_{g_0} h_0 = 0$ reads

$$\Phi^*(d\text{Vol}_g) = d\text{Vol}_{\Phi^*g} = d\text{Vol}_{g_0}(1 + \mathcal{O}(r^{-5})). \quad (9)$$

It follows that for all $\rho_2 > \rho_1 \gg 1$, in view of changing of variables, we have

$$\begin{aligned} g(\rho_2) - g(\rho_1) &= \text{Vol}_g(\Omega_{\rho_2} \setminus \Omega_{\rho_1}) - \text{Vol}_{g_0}((B_{\rho_2} \setminus B_{\rho_1})(0)/\Gamma) \\ &= \int_{B_{\rho_2} \setminus B_{\rho_1}} (\Phi^*(d\text{Vol}_g) - d\text{Vol}_{g_0}). \end{aligned}$$

Using $\Phi^*(d\text{Vol}_g) - d\text{Vol}_{g_0} = \mathcal{O}(r^{-5}) d\text{Vol}_{g_0}$, we obtain

$$|g(\rho_2) - g(\rho_1)| \leq C \int_{\rho_1}^{\rho_2} r^{-5} r^3 dr \leq C \rho_1^{-1}.$$

Hence $g(\rho)$ is a Cauchy function at infinity, and therefore admits a finite limit as $\rho \rightarrow \infty$. This proves that the renormalized volume is well defined.

3.3 A potential-theoretic expression for the renormalized volume

Once the CMC gauge has been constructed, the renormalized volume is already well defined as a geometric limit of volume differences. The next task is to understand its sign. For this purpose, it is not convenient to work directly with the definition in terms of volume differences. That definition is expressed through large boundary hypersurfaces, whereas the Ricci-flat condition is an equation on the interior. It is therefore natural to replace the volume defect by another quantity that carries the same information but is easier to relate to $\text{Ric}(g) = 0$.

The model for this comes from the flat cone $(\mathbb{R}^4/\Gamma, g_0)$. There the function r^2 is the natural potential for the radial scaling field $2r\partial_r$: one has

$$\Delta_{g_0} r^2 = 8, \quad \text{Hess}_{g_0}^0(r^2) = 0.$$

Thus r^2 encodes the exact homothetic structure of the cone. On a general Ricci-flat ALE space one cannot expect this scaling structure to persist exactly, but one may try to recover it asymptotically by solving

$$\Delta_g u = 8$$

with $u \sim r^2$ at infinity.

At this point one should explain why the more precise expansion

$$u = r^2 + br^{-2} + O_\varepsilon(r^{-3+\varepsilon}) \quad (10)$$

is the natural one to look for. In the good asymptotic gauge, we have

$$\Delta_{\Phi^*g}(r^2) = 8 + O(r^{-5}).$$

Hence, if we write $u = r^2 + v$ and require $\Delta_g u = 8$, then the correction term v should solve a Poisson equation with right-hand side of order $O(r^{-5})$. On a 4-dimensional ALE space, the Green function decays like r^{-2} , so the leading decaying term of such a solution is expected to be a multiple of r^{-2} . Thus the expansion in (10) is exactly the one suggested by the elliptic behavior of the equation.

There is also a simple geometric reason to single out the coefficient b . Later we will study a boundary integral over Σ_ρ . The derivative of br^{-2} is of order ρ^{-3} , and after integration over a hypersurface of area of order ρ^3 , this produces a term of order one. So this is the first decaying term in the expansion of u that may survive in the limit. By contrast, faster decaying terms contribute only $o(1)$. This suggests that b is the first coefficient that may be related to the renormalized volume.

We now construct such a function. Since $\Delta_{\Phi^*g}(r^2) = 8 + O(r^{-5})$, the function r^2 is already an approximate solution of $\Delta_g u = 8$ on the end. Extend the function $\Phi_* r^2 = r^2 \circ \Phi^{-1}$ from $M \setminus K$ to a smooth function u_0 on all of M , and set $f := 8 - \Delta_g u_0$, then $\Phi^* f = O(r^{-5})$. Standard weighted elliptic theory on ALE spaces produces a solution $\bar{u}$ of

$$\Delta_g \bar{u} = f$$

such that

$$\Phi^* \bar{u} = br^{-2} + O_\varepsilon(r^{-3+\varepsilon}).$$

Therefore

$$u := u_0 + \bar{u} = r^2 + br^{-2} + O_\varepsilon(r^{-3+\varepsilon}) \quad (11)$$

satisfies $\Delta_g u = 8$. And the discussion above shows why the coefficient b should be encoded the renormalized volume.

Integrating $\Delta_g u = 8$ over Ω_ρ , we obtain

$$8 \text{Vol}_g(\Omega_\rho) = \int_{\partial\Omega_\rho} \partial_\nu u (\nu \lrcorner d \text{Vol}_g), \quad (12)$$

where ν denotes the exterior unit normal to $\partial\Omega_\rho$.

Before expanding $\partial_\nu u$, we first record the size of $dr(\nu)$ on the boundary. Since $\partial\Omega_\rho = \Sigma_\rho$ is the level set $\{r = \rho\}$ in the CMC gauge, its exterior unit normal is $\nu = \frac{\nabla_g r}{|\nabla_g r|_g}$. Hence

$$dr(\nu) = \frac{dr(\nabla_g r)}{|\nabla_g r|_g} = |\nabla_g r|_g = \sqrt{g^{-1}(dr, dr)} = \sqrt{g^{rr}}.$$

Now write $g = g_0 + h_0 + h'$, $h_0 = O(r^{-4})$, $h' = O(r^{-5})$. Since $g_0 = dr^2 + r^2\gamma$, we have $(g_0)^{rr} = 1$. Expanding the inverse metric to first order gives

$$g^{rr} = 1 - h_0(\partial_r, \partial_r) + O(r^{-5}).$$

In particular,

$$dr(\nu) = 1 + O(\rho^{-4}),$$

so that $dr(\nu) - 1 = O(\rho^{-4})$. With this result, we can expand $\partial_\nu u$. Using (11), we have on the boundary $\partial\Omega_\rho = \Sigma_\rho$,

$$\begin{aligned}\partial_\nu u &= 2\rho dr(\nu) - 2b\rho^{-3}dr(\nu) + O_\varepsilon(\rho^{-4+\varepsilon}) \\ &= 2\rho - 2b\rho^{-3} + 2\rho(dr(\nu) - 1) + O_\varepsilon(\rho^{-4+\varepsilon}).\end{aligned}\tag{13}$$

At this point one uses the good gauge once more. From the more precise expansion above,

$$dr(\nu) - 1 = -\frac{1}{2}h_0(\partial_r, \partial_r) + O(r^{-5}).$$

Now the condition $\partial_r \lrcorner h_0 = 0$ implies $h_0(\partial_r, \partial_r) = 0$, and therefore in fact

$$dr(\nu) - 1 = O(\rho^{-5}).$$

Substituting this back into the expansion of $\partial_\nu u$ (13), we obtain

$$\partial_\nu u = 2\rho - 2b\rho^{-3} + O_\varepsilon(\rho^{-4+\varepsilon}).$$

On the other hand, from the expansion of the volume form (9), it follows that on Σ_ρ ,

$$\begin{aligned}\nu \lrcorner d\text{Vol}_g &= (1 + O(\rho^{-5})) \nu \lrcorner d\text{Vol}_{g_0} \\ &= (1 + O(\rho^{-5})) dr(\nu) (\partial_r \lrcorner d\text{Vol}_{g_0}) \\ &= (1 + O(\rho^{-5})) (\partial_r \lrcorner d\text{Vol}_{g_0}).\end{aligned}$$

Therefore the integrand in (12) can be expanded as

$$\partial_\nu u (\nu \lrcorner d\text{Vol}_g) = \left(2\rho - 2b\rho^{-3} + O_\varepsilon(\rho^{-4+\varepsilon})\right) (\partial_r \lrcorner d\text{Vol}_{g_0}).$$

Integrating over $\partial\Omega_\rho = \Sigma_\rho$, we obtain

$$\begin{aligned}8\text{Vol}_g(\Omega_\rho) &= \int_{\partial B_\rho(0)/\Gamma} \left(2\rho - 2b\rho^{-3} + O_\varepsilon(\rho^{-4+\varepsilon})\right) (\partial_r \lrcorner d\text{Vol}_{g_0}) \\ &= 2|S^3/\Gamma| \rho^4 - 2b|S^3/\Gamma| + O_\varepsilon(\rho^{-1+\varepsilon}).\end{aligned}$$

Recalling that $\text{Vol}_{g_0}(B_\rho(0)/\Gamma) = \frac{1}{4}|S^3/\Gamma| \rho^4$, we conclude that

$$8\text{Vol}_g(\Omega_\rho) = 8\text{Vol}_{g_0}(B_\rho(0)/\Gamma) - 2b|S^3/\Gamma| + O_\varepsilon(\rho^{-1+\varepsilon}).\tag{14}$$

Passing to the limit as $\rho \rightarrow \infty$ and comparing with the definition of the renormalized volume, we conclude that

$$b|S^3/\Gamma| = -4\mathcal{V}(M, g).\tag{15}$$

This is the key output of the present step. It shows that the renormalized volume can be recovered from the asymptotic expansion of the special solution u . In other words, the volume defect is encoded by the coefficient b , and the sign problem for $\mathcal{V}(M, g)$ is thereby reduced to a sign problem for this coefficient.

3.4 Nonpositivity of the renormalized volume

The identity (15) reduces the sign of the renormalized volume to the sign of the coefficient b . Thus the remaining task is to prove that $b \geq 0$.

The point is to place b into an integral identity whose right-hand side is manifestly nonnegative. The relevant tensor is the trace-free Hessian $\text{Hess}^0 u$. This choice is natural because $\text{Hess}^0 u$ measures the failure of ∇u to generate the homothetic structure of the flat cone: indeed, $\text{Hess}^0 u = 0$ if and only if ∇u is a conformal Killing field, and since $\Delta_g u = 8$ is constant, this is equivalent to saying that ∇u generates a one-parameter family of homotheties.

More importantly for the argument, $\text{Hess}^0 u$ satisfies a divergence identity. Indeed, since Hess can be composed as $\text{Hess}^0 u = \text{Hess} u - \frac{\Delta_g u}{4} g$, we have

$$\begin{aligned}\text{div}(\text{Hess}^0 u) &= \text{div}(\text{Hess} u) - \frac{1}{4} \text{div}((\Delta_g u)g) \\ &= \text{Ric}(du) + \frac{3}{4} d(\Delta_g u) = 0,\end{aligned}$$

since $\text{Ric}(g) = 0$ and $\Delta_g u = 8$ is constant.

To integrate by parts, set $T := \text{Hess}^0 u$, $X := T(\nabla u, \cdot)^\sharp$. Then

$$\text{div} X = \langle \text{div} T, du \rangle + \langle T, \text{Hess} u \rangle = \langle T, \text{Hess} u \rangle, \quad (16)$$

because $\text{div} T = 0$. Since $\text{Hess} u = T + \frac{\Delta_g u}{4} g$ and T is trace-free, we have

$$\langle T, \text{Hess} u \rangle = |T|^2 = |\text{Hess}^0 u|^2. \quad (17)$$

Therefore, combining (16) with (17) implies

$$\text{div} X = |\text{Hess}^0 u|^2.$$

Integrating over Ω_ρ and applying the divergence theorem gives

$$\begin{aligned}\int_{\Omega_\rho} |\text{Hess}^0 u|^2 d\text{Vol}_g &= \int_{\partial\Omega_\rho} g(X, \nu) (\nu \lrcorner d\text{Vol}_g) \\ &= \int_{\partial\Omega_\rho} (\text{Hess}^0 u)(\nu, \nabla u) (\nu \lrcorner d\text{Vol}_g),\end{aligned}$$

where in the last step we used the definition of X and the symmetry of $\text{Hess}^0 u$. Hence

$$\int_{\partial\Omega_\rho} (\text{Hess}^0 u)(\nu, \nabla u) (\nu \lrcorner d\text{Vol}_g) = \int_{\Omega_\rho} |\text{Hess}^0 u|^2 d\text{Vol}_g. \quad (18)$$

The right-hand side is nonnegative. Therefore, if the boundary term on the left can be identified asymptotically with a positive constant multiple of b , then b must also be nonnegative.

This is exactly what the final boundary expansion shows: after expanding $\text{Hess}^0 u$ and using the good-gauge conditions, the left-hand side of (18) is asymptotic to $12b|S^3/\Gamma|$. Passing to the limit in (18) therefore yields

$$12b|S^3/\Gamma| = \int_M |\text{Hess}^0 u|^2 d\text{Vol}_g \geq 0,$$

and hence $b \geq 0$.

It remains to compute the boundary term in (18). From this point on, we work on the asymptotic model $(\mathbb{R}^4 \setminus B_R(0))/\Gamma$ via the CMC gauge chart. After renaming the new chart as Φ , we suppress the pull-back notation and write Φ^*g simply as g . In particular, the radius function r , the vector field ∂_r , and the metric g_0 all refer to the Euclidean cone coordinates on the model end, and the hypersurface Σ_ρ is identified with the coordinate sphere $\{r = \rho\}$.

Since $\dim M = 4$ and $\Delta_g u = 8$, the trace-free Hessian can be written in local coordinates as

$$(\text{Hess}^0 u)_{ij} = (\text{Hess} u)_{ij} - \frac{\Delta_g u}{4} g_{ij} = \partial_i \partial_j u - \Gamma_{ij}^k \partial_k u - 2g_{ij}. \quad (19)$$

We now expand each term by using the asymptotic formulas for g and u .

From (11) we obtain

$$\begin{aligned}\partial_k u &= 2x_k + O(r^{-3}), \\ \partial_i \partial_j u &= 2\delta_{ij} - \frac{2b}{r^4} \left(\delta_{ij} - 4 \frac{x_i x_j}{r^2} \right) + O_\varepsilon(r^{-5+\varepsilon}).\end{aligned}$$

On the other hand, from $g_{ij} = \delta_{ij} + h_{0,ij} + O(r^{-5})$ we get

$$\Gamma_{ij}^k = \frac{1}{2}(\partial_i h_{0,jk} + \partial_j h_{0,ik} - \partial_k h_{0,ij}) + O(r^{-6}).$$

Substituting these expansions into the general formula (19) for $(\text{Hess}^0 u)_{ij}$, the terms $2\delta_{ij}$ and $-2\delta_{ij}$ cancel, and we obtain

$$\begin{aligned} (\text{Hess}^0 u)_{ij} = & -\frac{2b}{r^4} \left(\delta_{ij} - 4\frac{x_i x_j}{r^2} \right) \\ & - (\partial_i h_{0,jk} + \partial_j h_{0,ik} - \partial_k h_{0,ij}) x_k - 2h_{0,ij} + O_\varepsilon(r^{-5+\varepsilon}). \end{aligned} \quad (20)$$

We next evaluate this on the boundary term. From the previous estimates, on $\partial\Omega_\rho = \Sigma_\rho$ we have

$$\nu = \partial_r + O(\rho^{-4}), \quad \nabla u = 2\rho \partial_r + O(\rho^{-3}), \quad \nu \lrcorner d\text{Vol}_g = (1 + O(\rho^{-5}))(\partial_r \lrcorner d\text{Vol}_{g_0}).$$

Hence

$$\begin{aligned} (\text{Hess}^0 u)(\nu, \nabla u)(\nu \lrcorner d\text{Vol}_g) &= (\text{Hess}^0 u)(\partial_r + O(\rho^{-4}), 2\rho \partial_r + O(\rho^{-3}))(1 + O(\rho^{-5}))(\partial_r \lrcorner d\text{Vol}_{g_0}) \\ &= \left(2\rho (\text{Hess}^0 u)(\partial_r, \partial_r) + O(\rho^{-7}) \right) (\partial_r \lrcorner d\text{Vol}_{g_0}). \end{aligned} \quad (21)$$

It therefore remains to compute $(\text{Hess}^0 u)(\partial_r, \partial_r)$. For the part coming from br^{-2} , we contract the term $-\frac{2b}{r^4}(\delta_{ij} - 4\frac{x_i x_j}{r^2})$ in (20) with $\partial_r^i \partial_r^j$, where $(\partial_r)^i = x_i/r$. This gives

$$-\frac{2b}{r^4} \left(\delta_{ij} - 4\frac{x_i x_j}{r^2} \right) (\partial_r)^i (\partial_r)^j = -\frac{2b}{r^4} (1 - 4) = \frac{6b}{r^4},$$

and then

$$\begin{aligned} (\text{Hess}^0 u)(\partial_r, \partial_r) &= \frac{6b}{r^4} - \frac{1}{r^2} x_i x_j x_k (\partial_i h_{0,jk} + \partial_j h_{0,ik} - \partial_k h_{0,ij}) \\ &\quad - 2h_0(\partial_r, \partial_r) + O_\varepsilon(r^{-5+\varepsilon}). \end{aligned}$$

Thus, on $r = \rho$, (21) reads

$$\begin{aligned} (\text{Hess}^0 u)(\nu, \nabla u)(\nu \lrcorner d\text{Vol}_g) &= \left(12b\rho^{-3} - 2x_i x_j x_k (\partial_i h_{0,jk} + \partial_j h_{0,ik} - \partial_k h_{0,ij}) \rho^{-1} \right. \\ &\quad \left. - 4h_0(\partial_r, \partial_r)\rho + O_\varepsilon(\rho^{-4+\varepsilon}) \right) (\partial_r \lrcorner d\text{Vol}_{g_0}). \end{aligned} \quad (22)$$

We now simplify the terms involving h_0 in (22). Set

$$S := x_i x_j x_k (\partial_i h_{0,jk} + \partial_j h_{0,ik} - \partial_k h_{0,ij}) = 2x_i x_j x_k \partial_i h_{0,jk} - x_i x_j x_k \partial_k h_{0,ij}.$$

Since the coefficients of h_0 are homogeneous of degree -4 , Euler's relation gives $x_\ell \partial_\ell h_{0,ij} = -4h_{0,ij}$. Therefore

$$S = 2(-4x_j x_k h_{0,jk}) - (-4x_i x_j h_{0,ij}) = -4x_i x_j h_{0,ij} = -4r^2 h_0(\partial_r, \partial_r).$$

But the good-gauge condition $\partial_r \lrcorner h_0 = 0$ implies

$$h_0(\partial_r, \partial_r) = 0.$$

Hence $S = 0$, and all terms involving h_0 in (22) disappear. We are left with

$$(\text{Hess}^0 u)(\nu, \nabla u)(\nu \lrcorner d\text{Vol}_g) = (12b\rho^{-3} + O_\varepsilon(\rho^{-4+\varepsilon}))(\partial_r \lrcorner d\text{Vol}_{g_0}). \quad (23)$$

Integrating (23) over $\partial\Omega_\rho$ and letting $\rho \rightarrow \infty$, the identity (18) becomes

$$12b|S^3/\Gamma| = \int_M |\text{Hess}^0 u|^2 d\text{Vol}_g.$$

Since the right-hand side is nonnegative, we conclude that

$$b \geq 0. \quad (24)$$

Combining this with (15), we conclude that

$$\mathcal{V}(M, g) \leq 0. \quad (25)$$

This proves the desired inequality. Moreover, equality holds if and only if $\text{Hess}^0 u = 0$, equivalently, if and only if (M, g) is isometric to the flat cone $(\mathbb{R}^4/\Gamma, g_0)$.

3.5 What the proof is really doing

Let me summarize the proof in one conceptual paragraph.

The cone $(\mathbb{R}^4/\Gamma, g_0)$ possesses the homothetic field $r\partial_r$, whose potential is r^2 . The Ricci-flat ALE metric g is close enough to the cone that one can solve $\Delta_g u = 8$ with $u \sim r^2$, but the correction term br^{-2} measures the failure of the cone scaling structure to persist exactly. The renormalized volume is then just another expression of the same obstruction. The sign comes from the fact that this obstruction is measured by the square norm of $\text{Hess}^0 u$, and therefore cannot be negative.

This is why the proof feels so coherent once one has identified the main line: it is not a collection of unrelated expansions, but a single conformal-scaling argument written in asymptotic coordinates.

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